

Understanding Stack and Heap

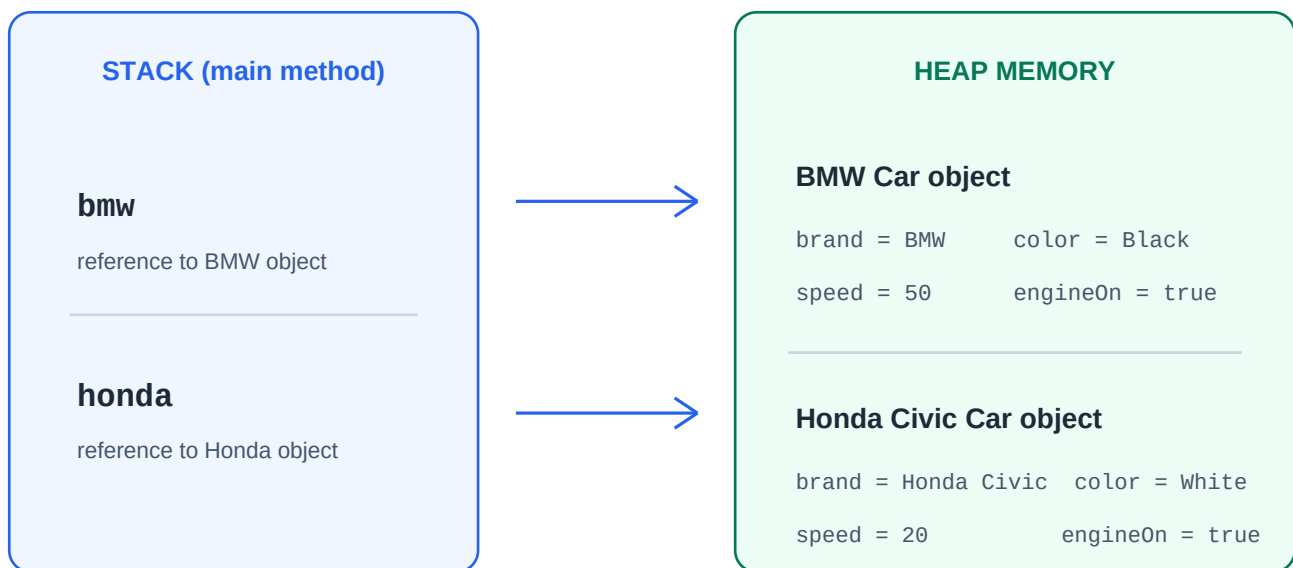
Khubaib Khan | B25110006066 | Section B

The important line

```
Car bmw = new Car();
```

When Java reaches this line, `bmw` is a reference variable and `new Car()` creates the actual object. The variable and the object are connected, but they are not the same thing.

What is stored where?



In simple words

The `bmw` and `honda` variables do not contain every detail of a car. Each one only stores a reference that points to its own object in heap memory. Since these are separate objects, the BMW and Honda Civic can have different colours, fuel levels and speeds. Stopping the BMW will not affect the Honda Civic.

Key takeaway

Same class, separate objects, separate state.